ENVS 193DS Midterm

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Set up

```r
# loading libraries
library(tidyverse)
library(janitor)
library(flextable)
library(rstatix)

# importing data
tussocks <- read_csv("2016ARFTussockHeight.csv")
temp <- read_csv("temp_2024.csv")
```

Problem 1. Understanding and critiquing written communication

a. Hypotheses

Null hypothesis (H_0): The difference in mean urchin gonad weight between heated and ambient treatments is equal to 0.

Alternative hypothesis (H_A): The difference in mean urchin gonad weight between heated and ambient treatments is *not* equal to 0.

b. Test type

Firstly, the researchers must not have known the population mean (average urchin gonad weight) or the standard deviation because they would have otherwise used the Z-test. Secondly, the data must have been normally distributed, and the variances must have been equal between group because they would have otherwise used the Welch's t-test.

c. Test summary components

The "t" in the citation represents the type of distribution we are looking at. We are using this distribution because the population standard deviation of urchin gonad weight is unknown.

The "53" represents the degrees of freedom. This is meant to represent the number of independent values in our data sample and is calculated by: $df = n_{heated} + n_{ambient} - 2$.

The value "-13.51" represents the test statistic obtained. This represents the divergence of our data from the central tendency of our null distribution.

The expression "p < 0.001" communicates that the p-value, or probability of this result under the null distribution, is less than 0.001. This can be interpreted to mean that it is so unlikely that our data could be obtained from our null distribution, that we are confident that it must come from another distribution.

d. Missing information

One component they could have included is describing the effect size (large, moderate, small) using Cohen's d, which suggests the extent to which the difference in mean gonad weight between heated and ambient groups diverges from zero. This helps to build a biological narrative because by emphasizing the magnitude of a certain treatment, we understand the urgency of similar conditions in real fisheries amidst climate change.

Another component that they could have included is the sample sizes for both heated and ambient treatment groups. This is crucial to addressing the hypothesis test to ensure transparency around the statistical power of the test.

Problem 2. Interpretation and communication

a. Hypotheses

Null hypothesis (H_0): The difference of mean crop value in Kern County between 2022 and 2023 is equal to 0.

Alternative hypothesis (H_A): The difference of mean crop value in Kern County between 2022 and 2023 is *not* equal to 0.

b. Satistical summary

Two pieces of information that are the same between the table and the provided test output are the t-value, a unitless value rounded to -1.5, and the p-value, another unitless value rounded to 0.14.

One piece of information that the test output provides that the researchers do not is the 95% confidence interval, which represents the range in which we are 95% confident in that it contains the mean difference in crop value between our 2022 and 2023 groups, and is bounded by $\pm$ one margin of error (test statistic at $\alpha = 0.05 \times$ standard error) around the estimate (sample mean difference). This means that we are 95% certain our mean difference in crop value between years could actually be anywhere from -$51.6 million to $7.7 million, but this is not actually communicated in that paper, as it would be simpler and correct to say that there was not a significant difference.

Another piece of information that the test output provides that the researchers do not is the degrees of freedom, which is a unitless value that represents the number of independent observations that are "free" to vary when calculating the mean difference in crop value between years, calculated by df = N - 1. This is to say that because we are calculating the mean difference of 28 independent observations of crop values, 27 of those crop values are free to vary but the 28th must be specific value in order to achieve that mean difference, but the researchers simply included the total number of observations N and skipped the inclusion of the simple calculation.

c. Writing a new statement

We find no significant difference in mean crop value between 2022 (n = 28) and 2023 (n = 28) (Paired t-test, t(27) = -1.5, p = 0.14, α = 0.05).

d. Statistical implications

Climate change is expected to bring more weather volatility, including phenomena like precipitation whiplash which can cause severe flooding. This has economic implications for farmers, whose crops may be threatened by climatic instability. However, based on a paired t-test of 28 major crops in Kern County, there was no significant difference in crop value between 2022 (a dry year) and 2023 (a wet year with floods), indicating that farmers, on average, have adapted to precipitation whiplash successfully. However, farmers may continue to prepare for climatic variability by practicing similar management strategies like fallowing fields, using deficit irrigation, and crop switching.

Problem 3. Reproducing an analysis

a. Variables

The response variable is tussock mound diameter, which is a continuous variable measured in cm. The predicting variable is the presence of fire, which is a binary categorical variable indicated as "burned"/"unburned".

The Microtopography subsection within the Methods section states that researchers measured the response variable with a caliper, measuring in two directions (N-S and E-W).

b. Hypotheses

Null hypothesis (H_0): The difference in mean tussock mound diameter between burned and unburned plots is equal to 0.

Alternative hypothesis (H_A): The difference in mean tussock mound diameter between burned and unburned plots is *not* equal to 0.

c. Cleaning and organinzing data

```
tussocks_clean <- tussocks |> # use tussocks data
  clean_names() |> # clean column names
  mutate(site = case_when(
    site == "Sev" ~ "Burned", # when "Sev" shows up in the site column, replace with "Burned"
    site == "Unb" ~ "Unburned")) |> # when "Unb" shows up in the site column, replace with "Unburned"
  select(site, average_diameter) # only select site and average_diameter columns
```

```
tussocks_sample <- slice_sample(tussocks_clean, n = 10) # create a random sample of 10 observations
print(tussocks_sample) # show sample
```

```
## # A tibble: 10 x 2
##    site     average_diameter
##    <chr>               <dbl>
```

```
##  1 Unburned              7.5
##  2 Unburned             25.1
##  3 Unburned             20.5
##  4 Burned               23.9
##  5 Burned               35.8
##  6 Burned               25.4
##  7 Unburned             22.9
##  8 Unburned             20.0
##  9 Unburned             15.4
## 10 Burned               24.3
```

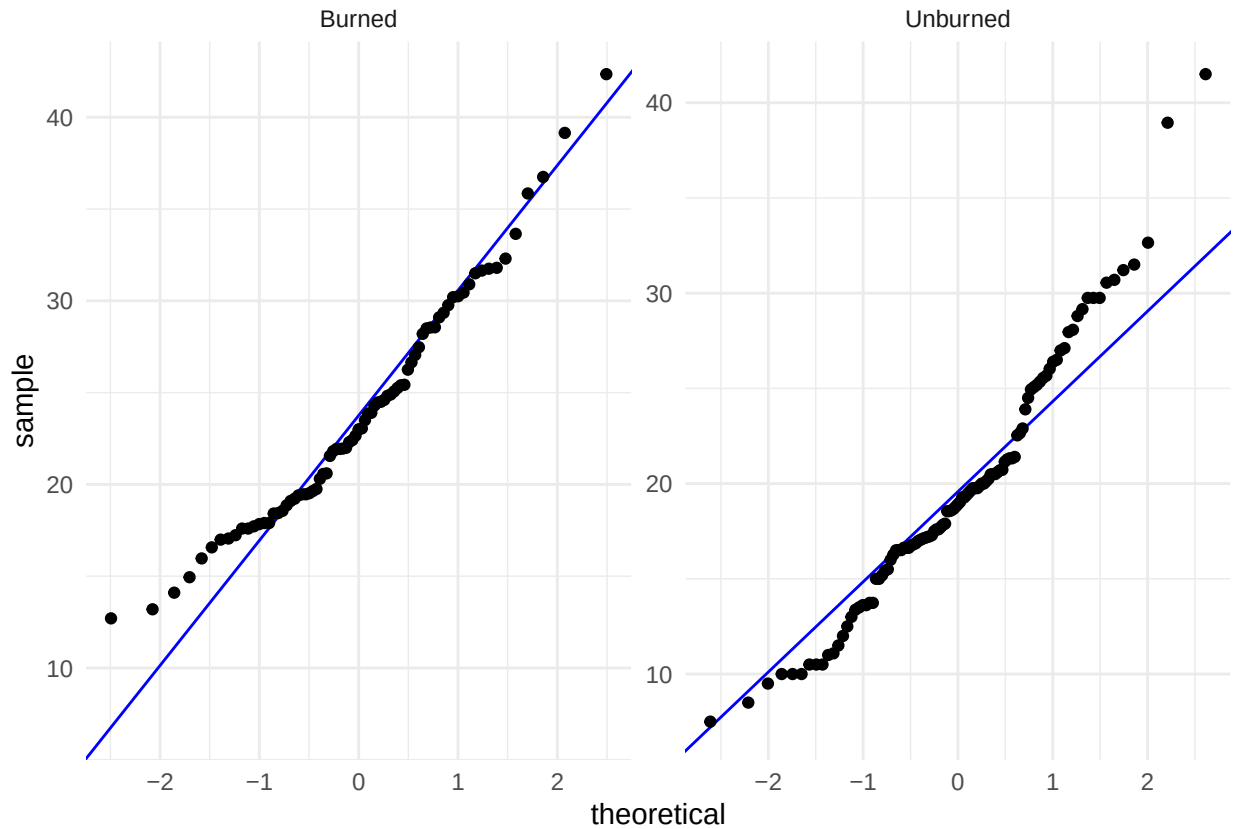
d. Summarizing and table displays

```
tussocks_summary <- tussocks_clean |> # use clean tussocks data
  group_by(site) |> # group by site
  summarise(mean = round(mean(average_diameter, na.rm = TRUE), digits = 1),
            # create a summary, save mean as mean diameters

            se = round(sd(average_diameter) / sqrt(length(average_diameter)), digits = 1))
            # save se as standard error of diameters

flextable(data = tussocks_summary) |>  # display a table of summary values
  set_header_labels(site = "Site", # Change column names
                    mean = "Mean",
                    se = "Standard error") |>
  autofit() # automatically adjust column size to fit content
```

Site	Mean	Standard error
Burned	23.8	0.7
Unburned	19.7	0.6

e. Check your assumptions

```
ggplot(data = tussocks_clean, # create a plot using tussocks data
       aes(sample = average_diameter)) + # set diameter as sample data
  geom_qq_line(color = "blue") + # add a qq line behind the qq plot, make it blue
  geom_qq() + # create a qq plot
  facet_wrap(~ site, # create different panels by site type
             scales = "free") +
  theme_minimal() # change the theme
```



We can see that our data points from burned and unburned sites roughly fall on the line representing the theoretical normal distribution, although we do see some deviations near the tail ends of our distribution. Regardless, we can use the Central Limit Theorem to validate that our sample means come from a normal distribution.

```
var.test(average_diameter ~ site, # check to see if the variance of diameter data is the
                                  # same between site groups
         data = tussocks_clean) # use tussock data
```

```
##
##  F test to compare two variances
##
## data:  average_diameter by site
## F = 0.92558, num df = 78, denom df = 110, p-value = 0.7221
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.6169823 1.4090774
## sample estimates:
## ratio of variances
##          0.9255843
```

We can also see that the variances of our two groups are the same (F test, $F(78) = 0.93$, $p = 0.72$, $\alpha = 0.05$).

f. Do a test

```
t.test(average_diameter ~ site, # compare mean mound diameters between burned and unburned sites
       data = tussocks_clean, # use tussocks data
       var.equal = T) # the variances of both groups are equal
```

```
##
##  Two Sample t-test
##
## data:  average_diameter by site
## t = 4.4663, df = 188, p-value = 1.371e-05
## alternative hypothesis: true difference in means between group Burned and group Unburned is not equal
## 95 percent confidence interval:
##  2.298075 5.934027
## sample estimates:
##   mean in group Burned mean in group Unburned
##               23.78632               19.67027
```

```
cohens_d(average_diameter ~ site, # find the effect size of site on diameters of tussock mounds
         data = tussocks_clean, # use tussock data
         var.equal = T) # variances are equal between groups
```

```
## # A tibble: 1 x 7
##   .y.              group1 group2   effsize    n1    n2 magnitude
## * <chr>            <chr>  <chr>      <dbl> <int> <int> <ord>
## 1 average_diameter Burned Unburned   0.657    79   111 moderate
```

g. Written communication

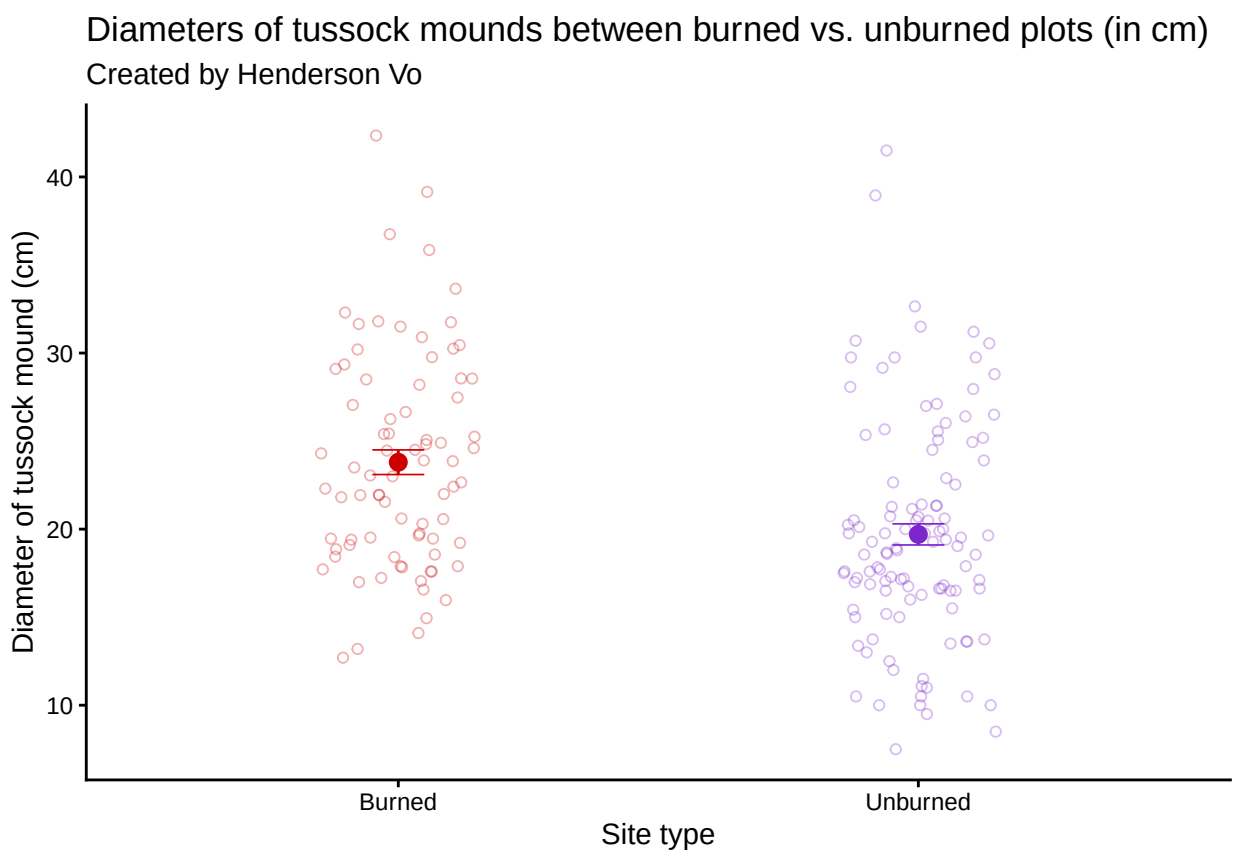
We have found a moderate (Cohen's D = 0.68) effect of burning on mean tussock mound diameter between burned (n = 79) and unburned (n = 111) sites (Student's t-test, t(188) = 4.47, p < 0.001, $\alpha = 0.05$).

On average, tussock mounds diameters were larger by 4 cm in burned sites (95% CI: [2.3, 5.9] cm), where the mean tussock mound diameter is 23.8 ± 0.7 cm (SE, n = 79) in burned sites and 19.7 ± 0.6 cm (SE, n = 111) in unburned sites.

h. Making a new figure

```
ggplot(data = tussocks_clean, # use tussocks data
       aes(x = site, # set x-axis to site
           y = average_diameter, # set y-axis to diameter
           color = site)) + # differentiate by site type

  geom_jitter(width = 0.15, # create a jitter plot, set width
              height = 0, # no height jitter
              alpha = 0.3, # set transparency of data points
              shape = 1) + # pick a different shape

  geom_errorbar(data = tussocks_summary,
```

```r
                aes(x = site, # set x-axis to site type
                      y = mean, # set point to mean
                      ymin = mean - se, # lower bound set to mean - standard error
                      ymax = mean + se), # upper bound set to mean + standard error
                width = 0.1, # set the width of error bar
                linewidth = 0.3) + # set the line thickness

geom_pointrange(data = tussocks_summary, # add a point range using summary data
                  aes(x = site, # set x-axis to site type
                      y = mean, # set point to mean
                      ymin = mean - se, # lower bound set to mean - standard error
                      ymax = mean + se)) + # upper bound set to mean + standard error

scale_color_manual(values = c(Burned = "red3", # set custom colors
                              Unburned = "purple3")) +

theme_classic() + # pick a cleaner theme

theme(legend.position = "none") + # remove the legend

labs(x = "Site type", # add labels to x- and y- axes, and add title and subtitle
     y = "Diameter of tussock mound (cm)",
     title = "Diameters of tussock mounds between burned vs. unburned plots (in cm)",
     subtitle = "Created by Henderson Vo")
```



i. Write a caption

Figure 1. Comparison of mean tussock mound diameter ± standard error (SE) at burned (n = 79) and unburned (n = 111) sites. Solid points represent the mean for each group, and the error bars show the standard error. Underlying points display individual observations of tussock mound diameter. This data comes from Rocha, A. 2021. "Tussock height and diameter in moist acidic tussock tundra at the site of the 2007 Anaktuvuk River fire scar, and nearby unburned tundra measured in 2016 ver 1." *Environmental Data Initiative*. https://doi.org/10.6073/pasta/1dccd3fdb3aa693f9c2b69a24f8306ed.

Problem 4. Cleaning, wrangling, and visualization

a. Initial cleaning, wrangling, and summarizing

```
temp_clean <- temp |>

  # 1. what package is this function from?
  # this is from the janitor package
  # 2. What changes after this function and how?
  # the column names change by setting all to lower case,
  # adding underscores between words, and removing special characters
  clean_names()|>

  # 3. what package is year() from?
  # year() comes from the tidyverse, specifically the lubridate package
  # 4. what is this combination of mutate() and year() doing?
  # this combination creates a year column in temp_clean by extracting
  # the years from the dates in temp
  mutate(year = year(date_local)) |>

  # 5. what package is filter() from?
  # filter() comes from the tidyverse, specifically the dplyr package
  # 6. what is being filtered in this functional call?
  # the sites are being filtered to only show certain site codes
  # 7. what sites and year(s) do you expect to be included
  # in the resulting data frame?
  # we should expect to see only sites NAPL, SCDI, and IVEE from the year 2024
  filter(site %in% c("NAPL", "SCDI", "IVEE") &
           year == "2024") |>

  # 8. what column is manipulated?
  # the site column
  # 9. what changes about the values in this column?
  # the site acronyms are changed to full names
  mutate(site = case_when(
    site == "IVEE" ~ "Isla Vista",
    site == "NAPL" ~ "Naples",
    site == "SCDI" ~ "Santa Cruz - Diablo"
  )) |>

  # 10. what is being calculated? and at what scale
  # (e.g. annual, monthly, daily)?
```

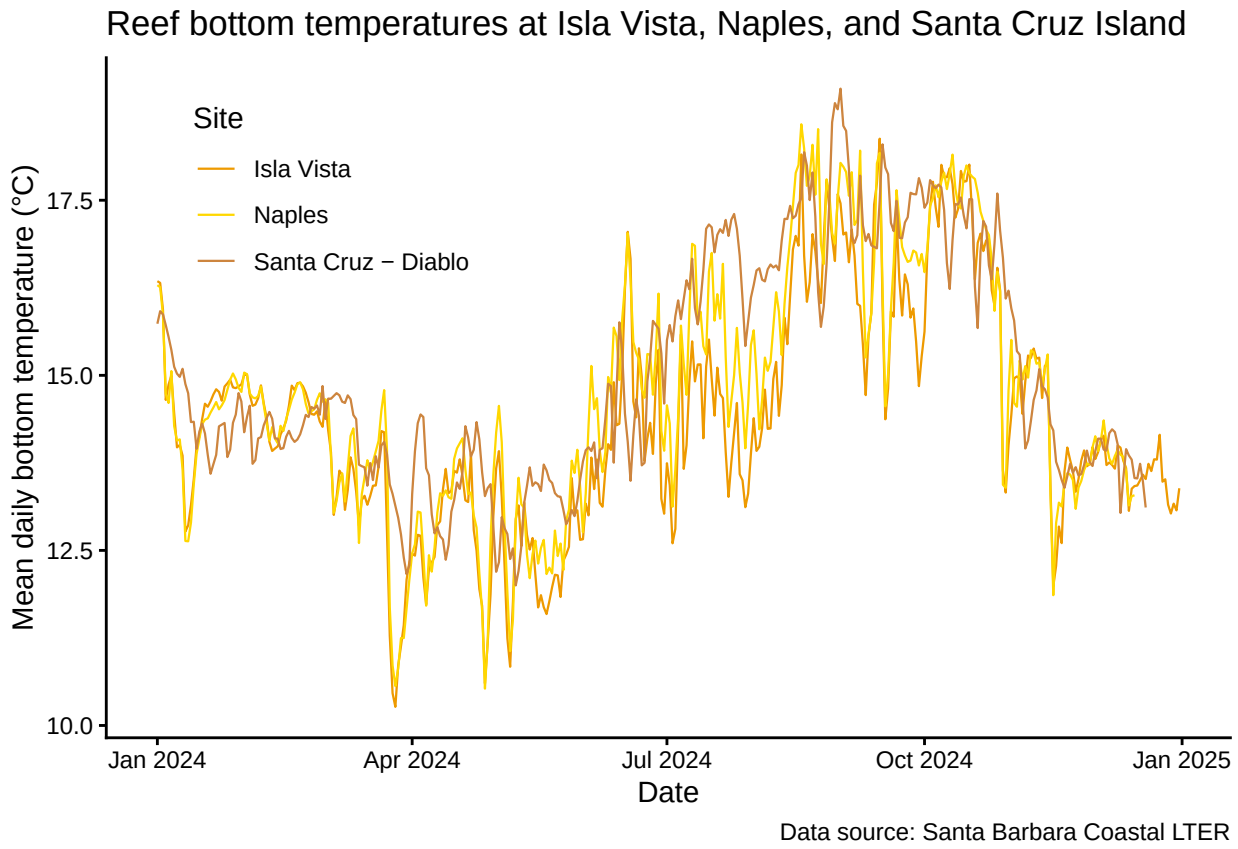
```r
  # the daily mean temperature is being calculated for each site
  # 11. give an example
  # Isla Vista on 2024-01-01 had a mean temperature of 16.3 degrees celcius
  group_by(site, date_local) |>
  summarize(mean_daily = mean(temp_c)) |>
  ungroup()
```

```r
slice_sample(temp_clean, n = 5) # display 5 random observations from temp_clean
```

```
## # A tibble: 5 x 3
##   site                date_local mean_daily
##   <chr>               <date>          <dbl>
## 1 Isla Vista          2024-02-07       14.9
## 2 Isla Vista          2024-07-24       13.8
## 3 Santa Cruz - Diablo 2024-03-14       13.7
## 4 Isla Vista          2024-10-17       18.0
## 5 Santa Cruz - Diablo 2024-06-12       13.8
```

b. Make the figure

```r
ggplot(data = temp_clean, # use temp data
       aes(x = date_local, # set x-axis to date
           y = mean_daily, # set y-axis to mean daily temperature
           color = site)) + # differentiate by site
  geom_line(linewidth = .4) + # add a line plot, control line thickness
  scale_color_manual(values = c("Isla Vista" = "orange2", # set colors for each site
                                "Naples" = "gold",
                                "Santa Cruz - Diablo" = "tan3")) +
  theme_classic() + # change the theme, removing gridlines
  theme(legend.position = "inside", # set the legend position
        legend.position.inside = c(0.2, 0.8)) +
  labs(x = "Date", # add labels to x- and y- axes, title, legend, and caption
       y = "Mean daily bottom temperature (\u00b0C)",
       title = "Reef bottom temperatures at Isla Vista, Naples, and Santa Cruz Island",
       caption = "Data source: Santa Barbara Coastal LTER",
       color = "Site")
```



c. Interpretation

The highest temperatures are recorded around September 2024 and the lowest temperatures are recorded around April 2024. We are able to tell because the y-axis indicates that temperature is higher as we travel up the axis. We can see that the lowest points on the three lines tend to fall around April 2024, and the highest points on the lines fall around September 2024.